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① Pattern Matching

A) With repetitions

Text: BANANA

Pattern: ANA

B A N A N A

A N A

Pattern occurs at position 2.

B A N A N A

A N A

Also occurs at position 4.

Recursive Approach :

Match (Text, Pattern, i, j)

If $j = \text{length}(\text{Pattern})$

return TRUE

If $i = \text{length}(\text{Text})$

return FALSE

If $\text{Text}[i] = \text{Pattern}[j]$

return Match (Text, Pattern, $i+1$, $j+1$)

Otherwise

return FALSE

Time complexity : $O(mn)$

B) Without repetitions

Text : ABCDEFGABC

Pattern : CDE

At position 1 :

ABC

$C \neq A$

At position 2 :

BCD

$C \neq B$

At position 3 :

CDE

$C = C$

$D = D$

$E = E$

$\therefore$ Pattern found at 3.

Recursive Approach :

Match(i, j)

If $j = m$

return TRUE

If $i = n$

return FALSE

If $\text{Text}[i] = \text{Pattern}[j]$

return Match($i+1, j+1$)

Otherwise

return Match($i+1, 0$)

Time complexity : $O(mn)$

② Overall Algorithm Complexity

* Initialization:

$$O(n^2 \log n)$$

* Main Computation:

$$O(n^3)$$

As these are sequential phases:

$$O(n^2 \log n) + O(n^3)$$

$$n^3 > n^2 \log n$$

$$\therefore \text{Overall complexity} = O(n^3)$$

③ Sum Algorithm

Algorithm sum(a, n) {

$s := 0.0;$

 for $i := 1$ to n do

$s := s + a[i];$

 return $s;$

}

We calculate number of steps.

Step 1: Initialization

$s := 0.0$

This executes 1 time. so cost = 1.

Step 2: Loop initialization

$i := 1$

Executes 1 time

Step 3: Loop Condition

$$i \leq n$$

It is checked $n+1$ times.

Step 4: Addition

$$S := S + a[i]$$

Runs exactly n times. so cost n .

Step 5: Increment

$$i := i + 1$$

Runs n times.

Step 6: Return

return S

Runs 1 time.

Total steps:

$$T(n) = 1 + 1 + (n+1) + n + n + 1 = 3n + 4$$

$$T(n) = 3n + 4$$

Ignoring constants:

$$T(n) = O(n)$$